

Reasoning questions in Unit 3: Energy resources
(Review questions from previous test papers)

1) Why 'bomb calorimeter' named as 'bomb'?

Answer:

In conducting a calorimetric test using a bomb calorimeter, the substance or object to be studied is placed inside the combustion crucible and ignited. The resulting reaction usually occurs so quickly that it resembles the explosion of a bomb-hence, the name "bomb calorimeter."

Bomb calorimeter consists of a strong steel vessel (called bomb) which can stand high pressure when the substance is burnt in it. Hence, it is called bomb calorimeter.

2) What is meant by water equivalent calorimeter?

Answer:

The amount of water that would absorb the same amount of heat as the calorimeter per degree temperature decrease.

3) What is specific heat of water?

Answer:

Specific heat of water (s) is the amount of heat energy required to increase the temperature of one kg of water by one degree centigrade.

4) Why the GCV value of a fuel is greater/higher than NCV value? (2M)

Answer:

Since while determining GCV, the amount of heat associated with steam (products of combustion) is taken into consideration, while in NCV determination, the products of combustion are let off to escape into atmosphere.

5) Which one does a bomb calorimeter measure either GCV or NCV? (2M)

Answer:

In a bomb calorimeter, GCV is measured since the experiment is performed at constant volume and the products of combustion are not allowed to escape but instead cooled to room temperature. Hence, GCV can be measured in a bomb calorimeter.

6) Justify: Reformation enhances octane rating of petrol? (2M)

Answer:

In reformation, straight chain hydrocarbons (low octane number) are converted into branched or aromatic hydrocarbons (high octane number). In other words, by simple structural modifications of the hydrocarbons in the petrol sample from straight chain hydrocarbons into branched or aromatic hydrocarbons, octane number (quality of the petrol) is increased. Therefore, reformation enhances octane rating of petrol.

7) Why cracking is required?

(2M)

Answer:

The main objective of cracking is to increase the quality of petrol in crude oil. After, cracking, only petrol fraction is subjected to further chemical treatment to improve the quality of petrol.

8) Give reason:

(2M)

Ethylene dibromide and dichloride's are used along with leaded anti-knocking agents to prevent knocking (or) TEL is used as anti-knocking agent along with ethyl bromide in IC engine?

Answer:

Tetraethyl lead (TEL) is added in the form of ethyl fluid and the petrol containing ethyl fluid is known as leaded (ethylated) petrol. Ethyl fluid contains 63% TEL, 26% dibromoethane (ethylene dibromide) and 9% dichloroethane (ethylene dichloride). Dibromoethane (ethylene dibromide) and dichloroethane (ethylene dichloride) are added to remove lead as volatile lead halides which escape with other gases through the exhaust.

9) Leaded anti-knocking agents cannot be used when catalytic converters are used (or) leaded petrol should not be used in automobiles, where catalytic converters are used.

(2M)

Answer:

A major advantage of unleaded petrol is it allows the use of catalytic converter to the exhaust in automobiles. The catalytic converter contains a rhodium catalyst which oxidizes the toxic gases such as CO and NO to CO₂ and N₂. It also oxidizes the unburnt hydrocarbons into CO₂ and H₂O.

Leaded petrol cannot be used in automobiles equipped with catalytic converter as the lead (Pb) poisons the catalyst.

Some points on Bomb Calorimeter (for understanding)

- 1) The Bomb Calorimeter was used to measure the cross calorific value of the solid and liquid samples.
- 2) Heat of combustion measured in an oxygen bomb calorimeter is the total heat available.
- 3) The bomb calorimeter is an instrument used to measure the heat of reaction at a fixed volume and the measured heat which is called the change of internal energy (ΔE) of hydrocarbons.
- 4) Benzoic acid (C₆H₅CO₂H) is often used for this purpose because it is a crystalline solid that can be obtained in high purity. The combustion of benzoic acid in a bomb calorimeter releases 26.38 kJ of heat per gram (i.e., its $\Delta H_{\text{comb}} = -26.38 \text{ kJ/g}$).
- 5) The characteristics of a good fuel should have the following:

1. **High calorific value:**

A fuel should possess high calorific value, since the amount of heat liberated and temperature attained thereby depends upon the calorific value of the fuel.

2. **Moderate ignition temperature:**

Ignition temperature is the lowest temperature to which the fuel must be pre-heated so that it starts burning smoothly. Low ignition temperature is dangerous for storage and transport of fuel, since it can cause fire hazards. On the other hand, high ignition temperature causes difficulty in igniting the fuel, but the fuel is safe

during storage, handling and transport. Hence, an ideal fuel should have moderate ignition temperature.

3. **Low moisture content:**

The moisture content of the fuel reduces the heating value and involves in a loss of money, because it is paid for at the same rate as the fuel. Hence, fuel should have low moisture content.